

Beyond Default: How to Calibrate Your Availability Profiles with Different Wind Turbines

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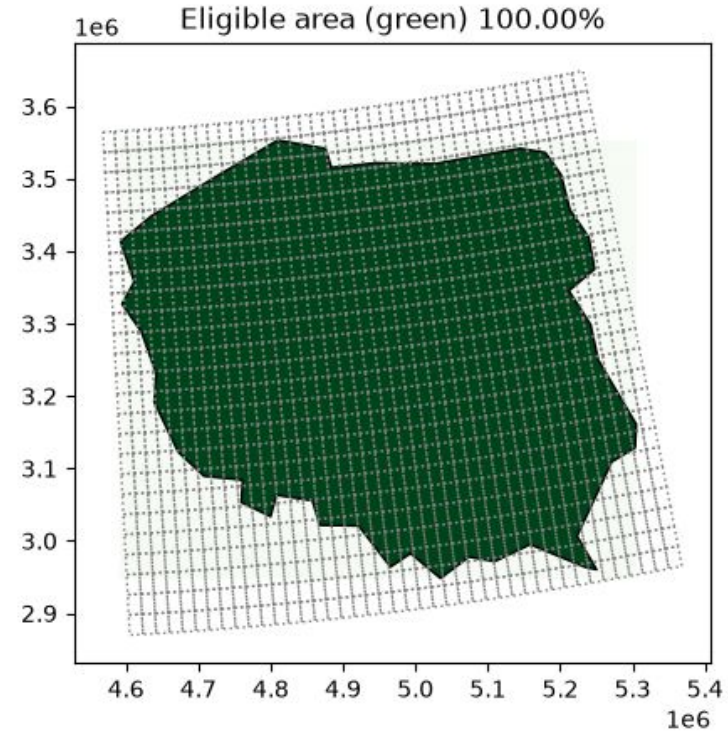
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Context I

**Modelling wind turbines
(in Poland) is complicated.**

0. Assumptions

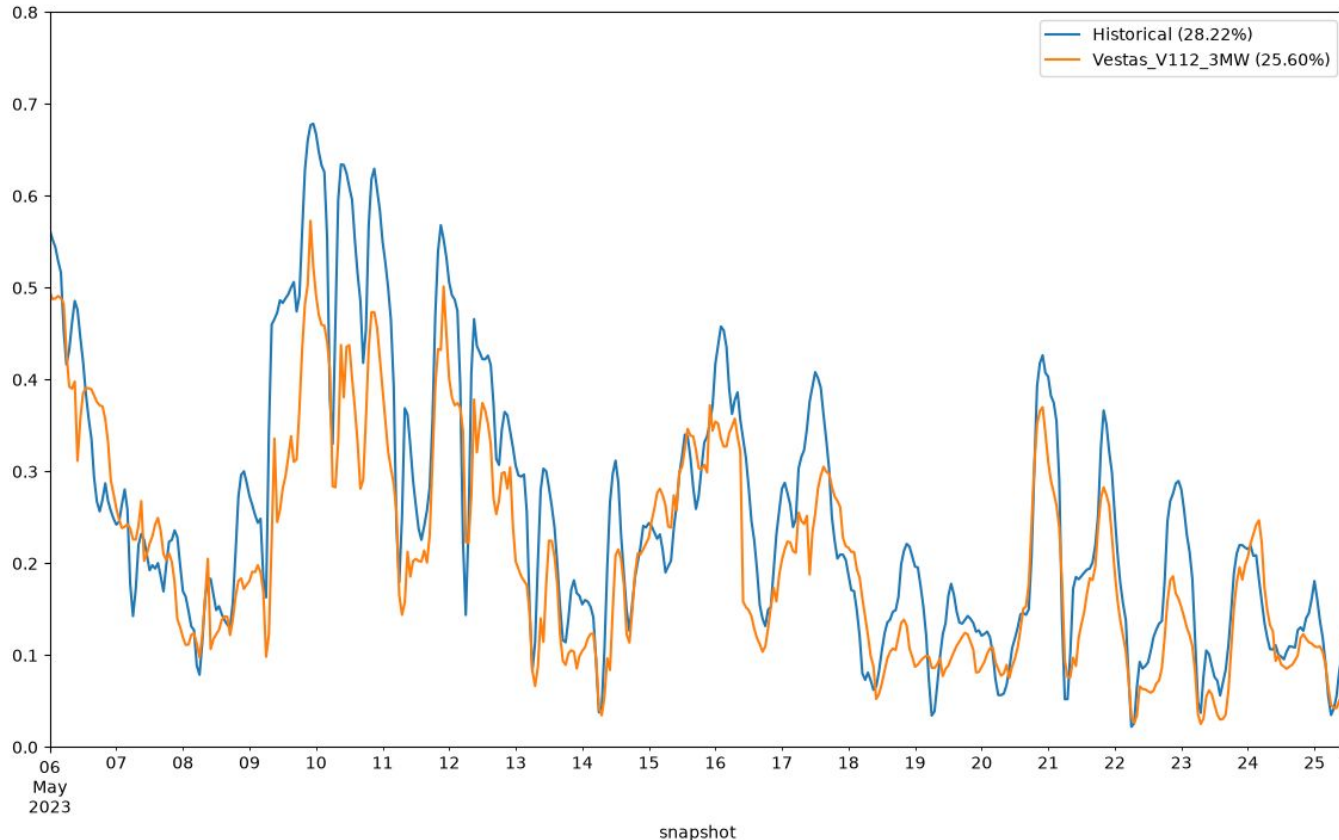
1. Downloaded cutout of a country
 - a. e.g. yearly weather from ERA5.
2. Historical generation and capacities are reliable
 - a. We combine generation from ENTSO-e and monthly updated capacities from ARE
3. No curtailment



Context II

Capacity factor is important, but that's not the whole story.

1. Default atlite configuration



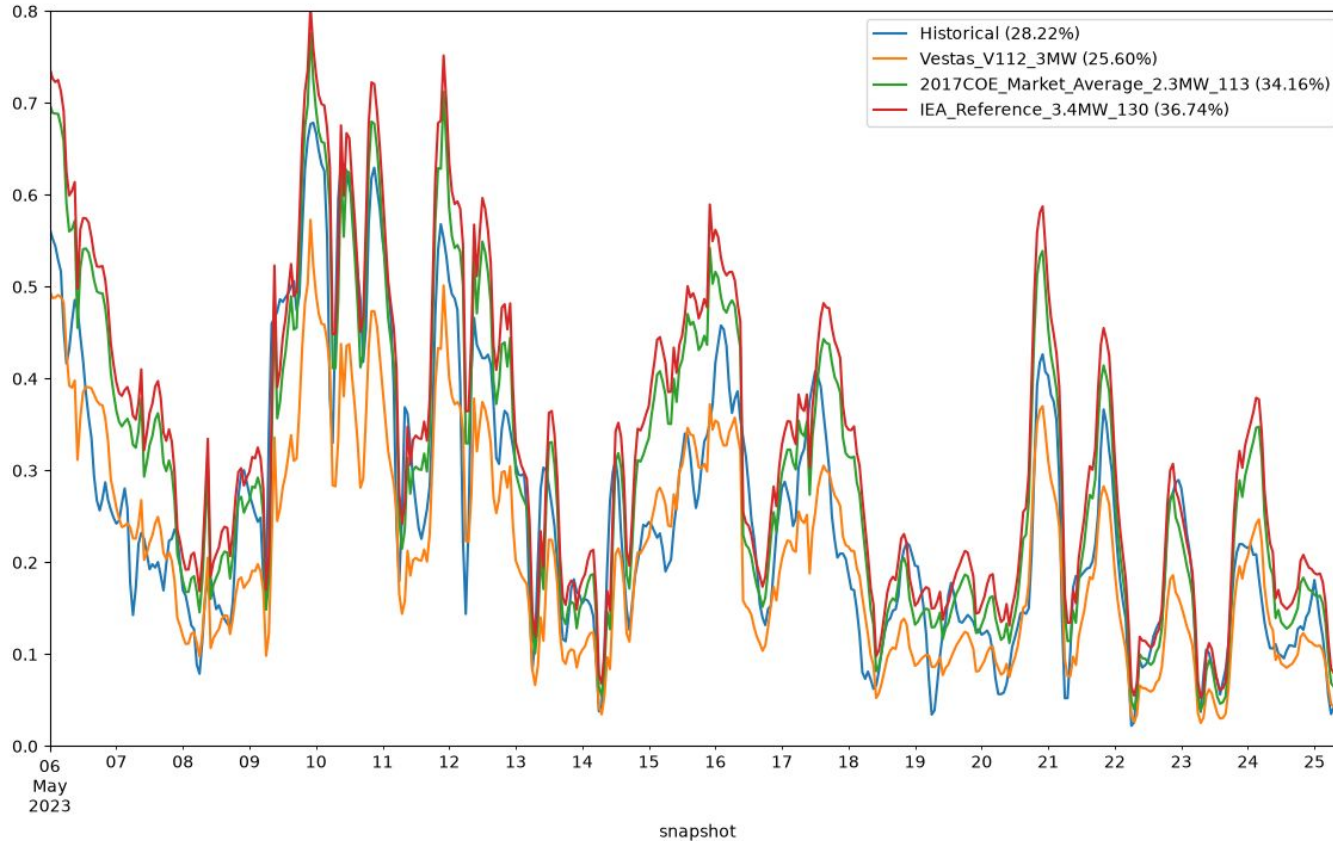
Vestas V112 3MW from a tutorial in atlite.

Used by default in PyPSA-Eur

CF is more or less in match, but peaks are underperforming

This was an issue in Instrat modelling

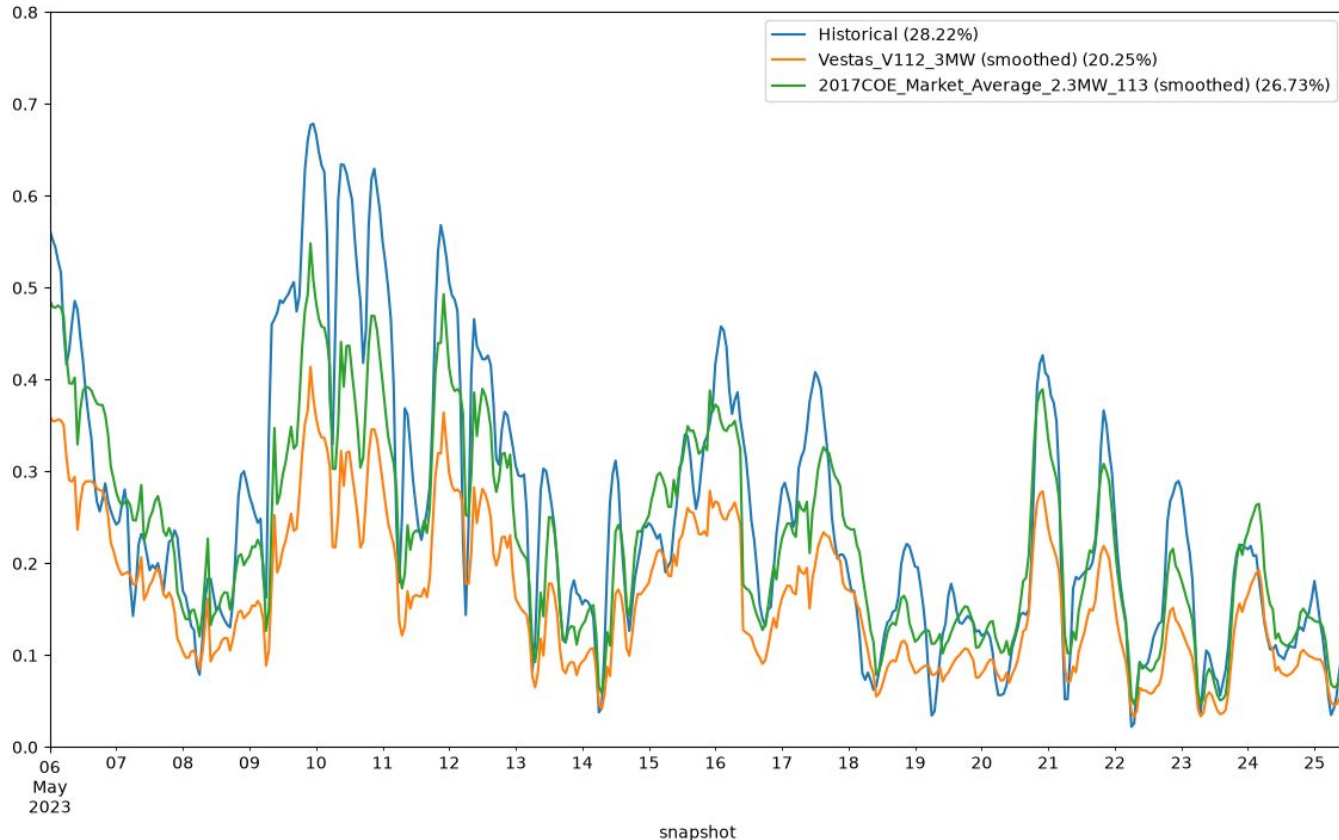
2. Change turbine



Atlite has over a dozen different turbine to choose from.

The NLR (formerly NREL) has even larger library.

3. Add smoothing

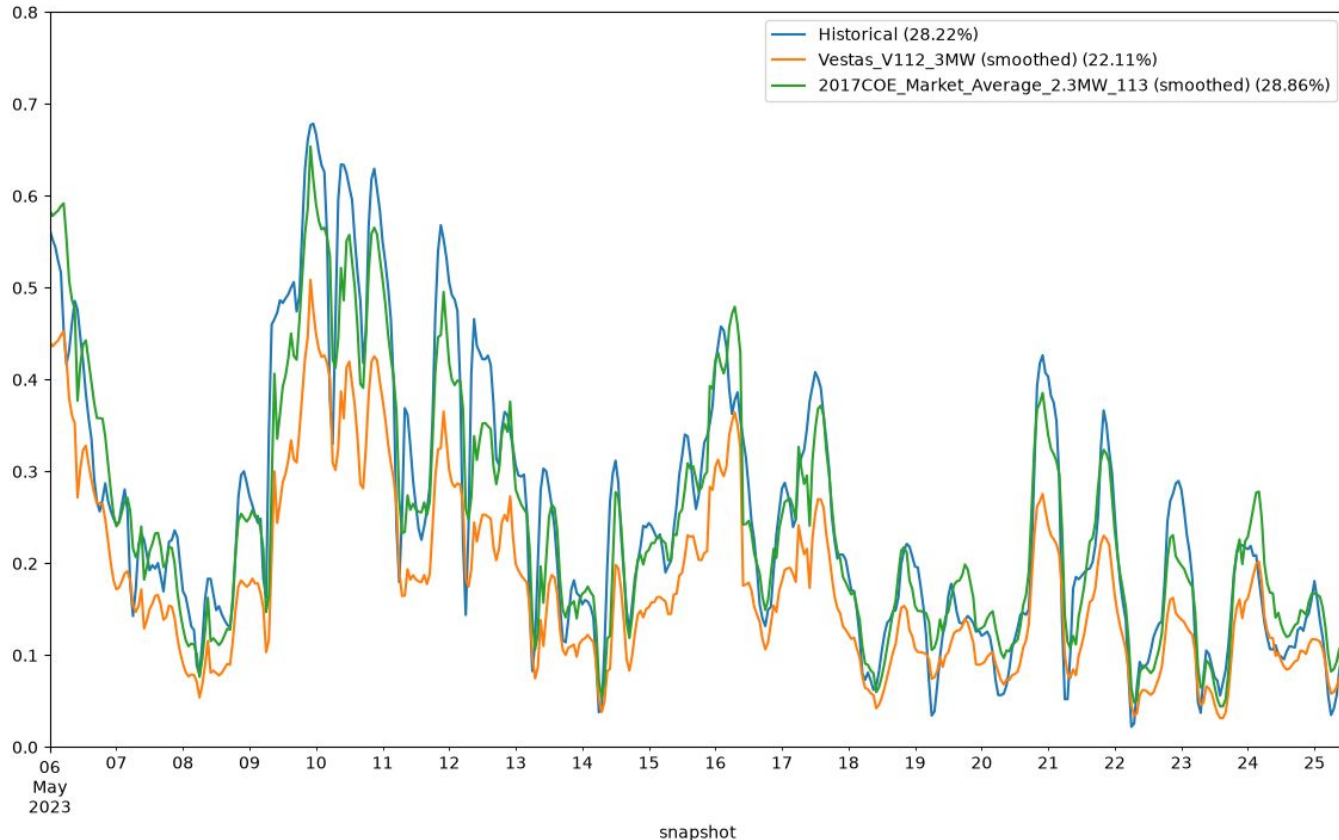


Gaussian filter over profile

Models technical outages and fluctuations of a fleet of turbines

Used by Andersen et al.
(<https://arxiv.org/abs/1409.3353>),
deployed in Atlite, possibly
renewables.ninja too

4. Create subdivisions with weights



Profile for each voivodeship
(province/prefecture)

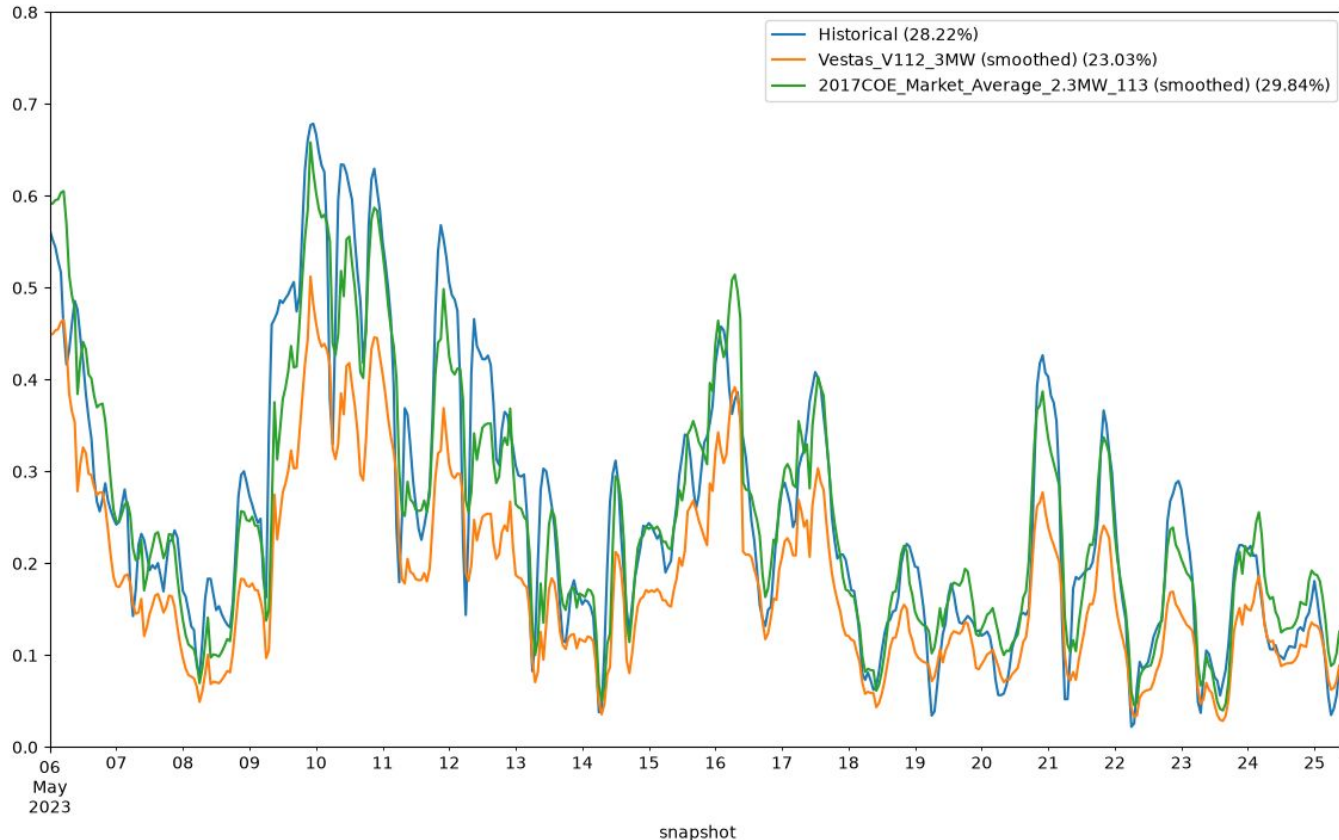
Country-wide weighted
average

Weights as a proportion of
current capacities.

More:

<https://instrat.pl/en/grids-and-res/>

5. Add geospatial features



Use existing turbines or potential ones.

You can exclude:

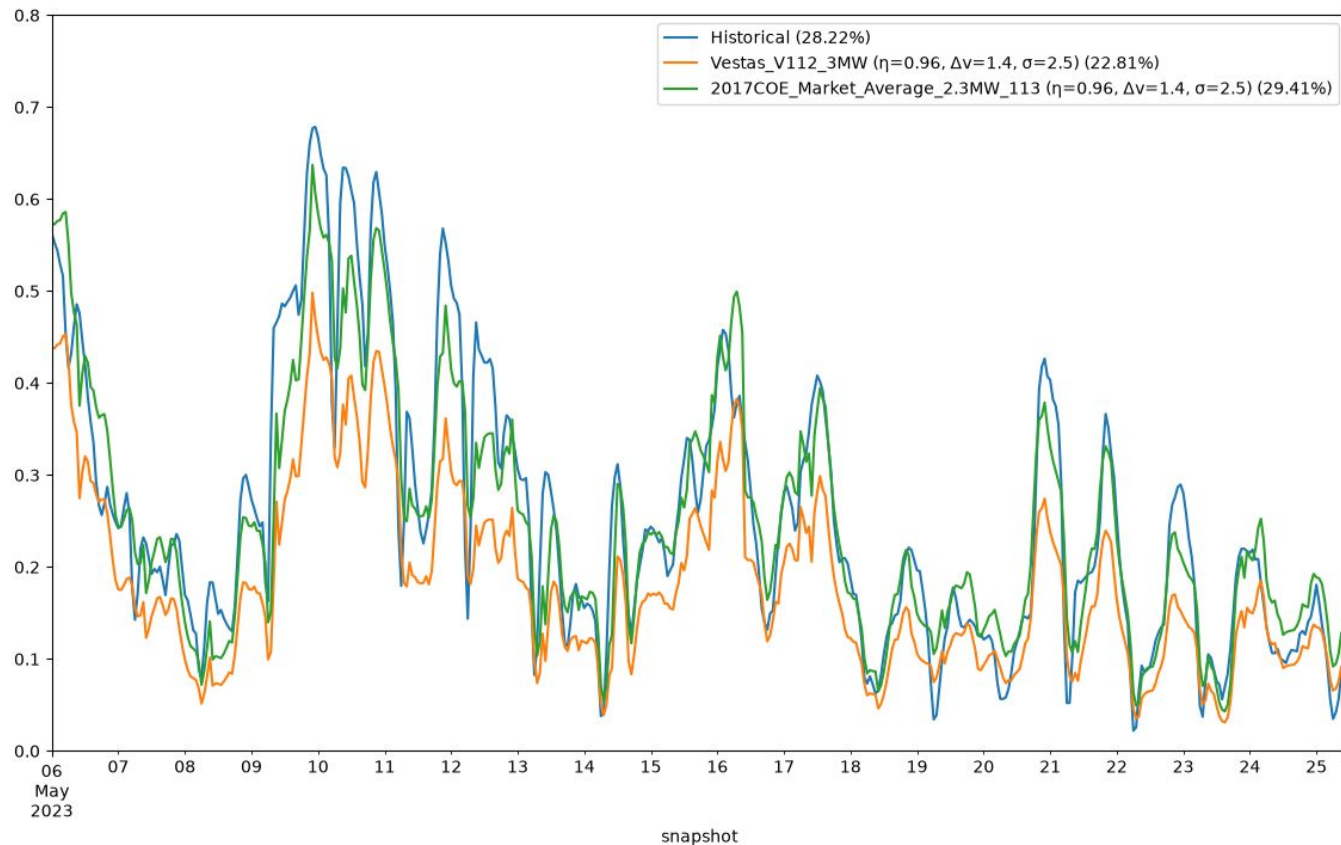
- difficult terrain
- zoning law requirements
- steep slopes

Change weights of weather cells according to exclusion masks

More:

<https://instrat.pl/ustawa-wiatrakowa-2024/>

6. Change settings of smoothing



There are 3 advanced settings in smoothing:

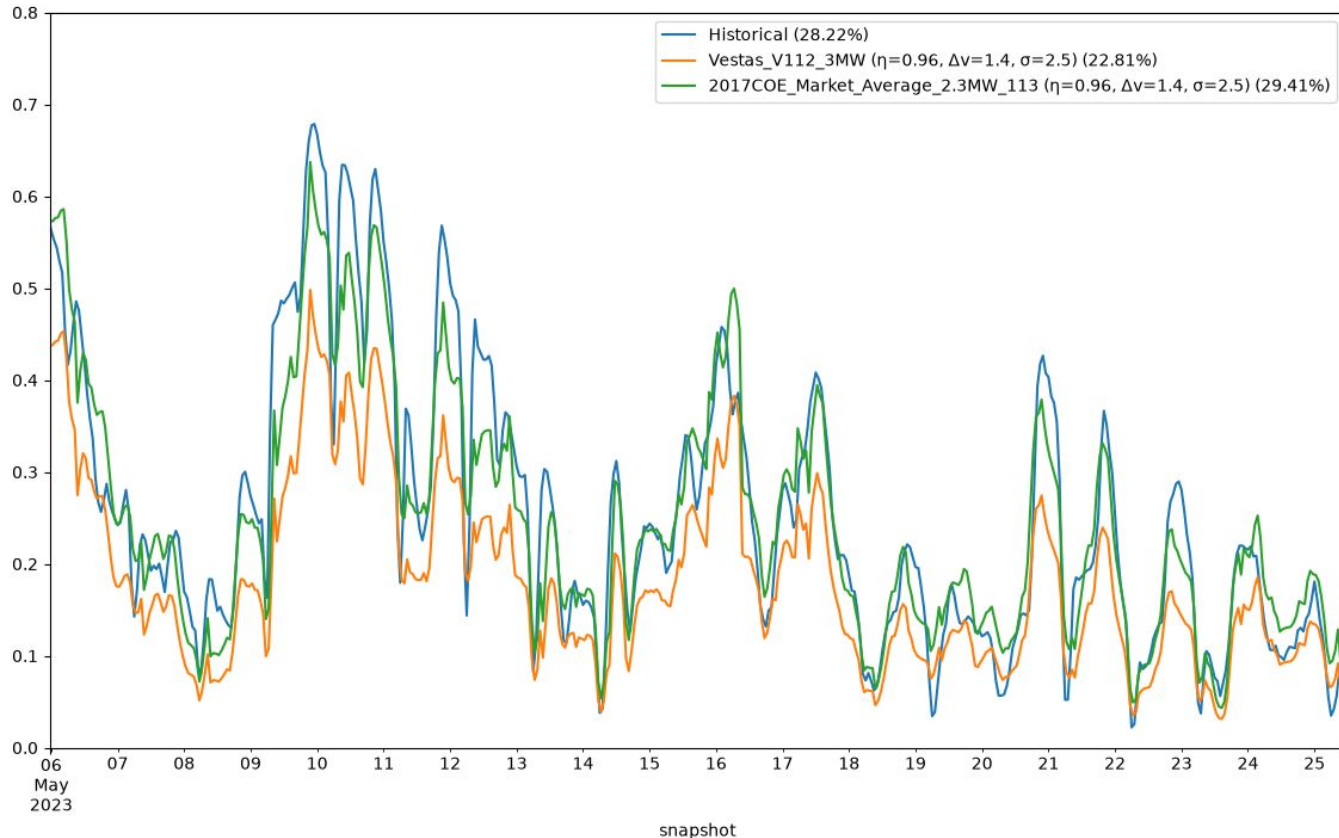
eta - fleet availability

Delta_v - offset of a filter

sigma - standard deviation

Be mindful of oversmoothing:
don't use settings that will cause electricity generation for zero wind speed.

7. Read ERA5 docs & comments in atlite's code



It's a timezone. Always.

17:00 sometimes means
16:01-17:00 and sometimes
17:00-17:59

Some data is average, other is
instantaneous

Atlite issue from 2021
proposed 30 min offset

Wrap-up

Check if the default works for you.

If not, make some changes.

If still not, make it custom.

If you have a resources and needs, go into the journey to understand what's behind the default.



Code used in this presentation is available on:
<https://github.com/instrat-pl/wind-wind>

Sources

Atlite

<https://github.com/PyPSA/atlite>

ARE (access through *energy.instrat.pl*)

<https://energy.instrat.pl/en/electrical-system/generation-capacity-are/>

ENSTO-E (access through *energy.instrat.pl*)

<https://energy.instrat.pl/en/electrical-system/electricity-production-entsoe/>

ERA5

<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels>

Geoportal (BDOT10k & outlines)

<https://www.geoportal.gov.pl/>

GLAES

<https://github.com/FZJ-IEK3-VSA/glaes>

NLR (formerly NREL)

<https://github.com/NatLabRockies/turbine-models>

Sonny

<https://sonny.4lima.de/>

URE

<https://www.ure.gov.pl/pl/oze/potencjal-krajowy-oze/8108.Instalacje-odnawialnych-zrodel-energii-stan-na-30-czerwca-2026-r.html>



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Thank you for your attention!

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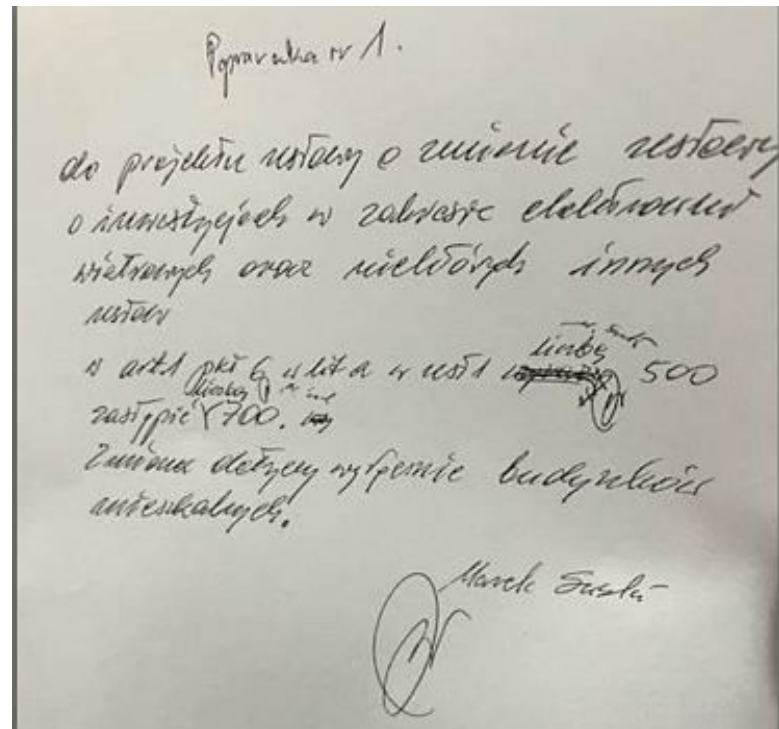
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4. Apply local insights

1. Repressive zoning laws.
 - a. 2016-2023: 10 x height of a turbine from the nearest house
 - b. 2023-currently: 700m if local zoning accepts that
2. Very long development time (up to 6-7 years)
3. Outdated turbines (2-3 MW)
 - a. It is easier to build a project developed pre-2016 than start a new one
4. Future growth would probably depend on the repowering
5. Obscure excel files on Energy Regulatory Office website contains reliable data about location of existing plants

Source



26.9%

**This is the
average historical capacity
factor of wind turbines in
Poland 2015-2025**